

# Abdullah

AI / ML Engineer

Python · Machine Learning · Deep Learning · LLM Agents · MLOps

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## Professional Summary

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Data Science student and AI developer who builds and deploys machine learning and agentic AI systems end to end: data ingestion, model training, API integration, cloud deployment, and monitoring. Experienced with Scikit-learn, TensorFlow/Keras, LangChain/LangGraph, FastAPI, Docker, and AWS(basic). **8th place worldwide** in the SmartReco Build Challenge 2026 (Krish Naik Academy Global Hackathon).

## Technical Skills

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- **Languages:** Python, SQL, C++ (basic)
- **ML / DL:** Scikit-learn, TensorFlow, Keras, CatBoost, Pandas, NumPy, feature engineering, hyperparameter tuning, model evaluation
- **Generative AI:** LangChain, LangGraph, RAG, ChromaDB, prompt engineering, structured outputs, LLM guardrails
- **MLOps & Deployment:** MLflow, DagsHub, Docker, GitHub Actions (CI/CD), AWS (S3, ECR, EC2)
- **Backend & APIs:** FastAPI, REST APIs, JWT authentication, SQLAlchemy, Alembic
- **Databases:** MongoDB Atlas, PostgreSQL, SQLite, ChromaDB (vector DB)
- **Concepts:** Data Structures & Algorithms, Statistics, NLP, Predictive Modeling, Model Monitoring (LangSmith)
- **Tools:** Git, GitHub, pytest, Streamlit, Jupyter

## Projects

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### AgentCare: Agentic AI for Patient Administration

*Python, FastAPI, LangGraph, SQLAlchemy, Docker*

[github.com/Abdullah-qazi-1/agentcare-ai-patient-coordination](https://github.com/Abdullah-qazi-1/agentcare-ai-patient-coordination)

- Built a multi-agent system with six specialized LangGraph agents and 23 tools that turns free-text patient requests into completed work: department routing, real appointment booking, document filing, and reminders.
- Designed a six-layer safety pipeline (deterministic scanner, LLM judgement, output and tool-argument inspection, NeMo Guardrails) so the AI never gives medical advice, backed by an audit trail and human-in-the-loop approvals that survive restarts.
- Prevented double-booking with atomic database slot-claiming, and cut LLM cost with deterministic fast paths, structured outputs, per-agent model tiering, and hard token budgets.
- Wrote 298 automated tests, red-team and routing-accuracy evals, a CI pipeline, and Docker packaging, with LangSmith/Logfire observability, and deployed a live demo.

### SmartReco: Behavioral AI Recommendation Agent

*8th Place Worldwide, SmartReco Build Challenge 2026*

Python, FastAPI, LangGraph, ChromaDB, RAG, APScheduler

- Built an e-learning recommendation platform that tracks learner behavior in real time and scores interest using recency decay (7-day half-life), dwell-time multipliers, and frequency boosts.
- Implemented a LangGraph agent (analyze, retrieve, evaluate, refine, generate) with RAG over ChromaDB, plus a grounding check that validates LLM output against the real course catalog to reduce hallucinations.
- Reduced unnecessary LLM calls with trigger-gated generation and response caching, and added proactive daily digests via email and Telegram.
- Hardened the system with a dual-write SQL and vector store with self-healing sync, graceful fallbacks when the LLM is unavailable, rate limiting, and fixes for privilege-escalation vulnerabilities, with smoke tests and a recommendation-quality eval script.

### Network Security: Phishing Website Detection (End-to-End MLOps)

*Scikit-learn, FastAPI, MongoDB, MLflow, Docker, AWS*

[github.com/Abdullah-qazi-1/End-to-End-Mloops-Project](https://github.com/Abdullah-qazi-1/End-to-End-Mloops-Project)

- Built a complete ML pipeline that ingests data from MongoDB Atlas and runs validation, transformation, and model training with automated hyperparameter tuning to classify phishing websites.
- Tracked experiments and model metrics with MLflow and DagsHub for reproducible model comparison and selection.
- Exposed the trained model through a FastAPI service for real-time batch predictions.

- Automated deployment with Docker and a GitHub Actions CI/CD pipeline that pushes images to AWS ECR and runs them on EC2, with S3 for artifact storage.

## Education

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**Bachelor of Science in Data Science**, Air University, Islamabad

Expected 2028

Relevant Coursework: Statistics, Machine Learning, Applied Data Engineering

## Certifications & Achievements

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- 8th Place Worldwide, SmartReco Build Challenge 2026 (Krish Naik Academy Global Hackathon)
- Complete Data Science, Machine Learning, Deep Learning & NLP Bootcamp
- Complete Python with Data Structures & Algorithms Bootcamp